

Suppose we are solving the system $Ax = b$. In practice, b is rarely exact.

It often contains small measurement errors (e.g. if we save 0.1 in float 32,

we probably save 0.100000012). We denote the error of b as Δb . So we have

Similarly, we have error for saving A

$$(A + \Delta A)(x + \Delta x) = \underbrace{b + \Delta b}_{\substack{\text{the real 'b' (red)} \\ \text{the b we save (blue)}}} + \underbrace{\Delta A x}_{\substack{\text{the additional term} \\ \text{because of the numerical approximation. (red)}}}$$

$\downarrow$
 the exact solution for $Ax = b$ (blue)

$$\Rightarrow Ax + A\Delta x + \Delta Ax + \Delta A\Delta x = b + \Delta b$$

$$\Rightarrow A\Delta x + \Delta Ax + \Delta A\Delta x = \Delta b$$

$\rightarrow$ drop this second order term
because it will be much smaller

$$\Rightarrow A\Delta x + \Delta Ax \approx \Delta b$$

$$\Rightarrow \Delta x = A^{-1}(\Delta b - \Delta Ax)$$

$$\Rightarrow \|\Delta x\| = \|A^{-1}(\Delta b - \Delta Ax)\| \quad \begin{matrix} \nearrow \text{matrix norm} \\ \|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} \end{matrix} \Rightarrow \|A\| \geq \frac{\|Ax\|}{\|x\|} \Rightarrow \|A\| \|x\| \geq \|Ax\|$$

$$\Rightarrow \|\Delta x\| \leq \|A^{-1}\| \|\Delta b - \Delta Ax\|$$

if the norm for vector here is L_2 norm,
then $\|A\| = \sigma_{\max}(A)$ (largest singular value)

$$\Rightarrow \|\Delta x\| \leq \|A^{-1}\| (\|\Delta b\| + \|\Delta Ax\|)$$

$$\Rightarrow \frac{\|\Delta x\|}{\|x\|} \leq \|A^{-1}\| \left(\frac{\|\Delta b\|}{\|b\|} + \|\Delta A\| \right)$$

$$\text{Since } Ax = b \Rightarrow \|Ax\| = \|b\| \Rightarrow \|b\| \leq \|A\| \|x\| \Rightarrow \frac{1}{\|x\|} \leq \frac{\|A\|}{\|b\|}$$

$$\Rightarrow \frac{\|\Delta x\|}{\|x\|} \leq \|A^{-1}\| \left(\frac{\|A\| \|\Delta b\|}{\|b\|} + \|\Delta A\| \right)$$

$$\Rightarrow \frac{\|\Delta x\|}{\|x\|} \leq \|A^{-1}\| \|A\| \left(\frac{\|\Delta b\|}{\|b\|} + \frac{\|\Delta A\|}{\|A\|} \right)$$

$$\Rightarrow \frac{\|\Delta x\|}{\|x\|} \leq K(A) \left(\frac{\|\Delta b\|}{\|b\|} + \frac{\|\Delta A\|}{\|A\|} \right)$$

Where $\kappa(A) = \|A\| \|A^{-1}\| = \frac{\sigma_{\max}}{\sigma_{\min}}$ is known as **conditional number**

What this means: if conditional value κ is small $\Rightarrow$ the error Δx is small relative to x
if conditional value κ is large $\Rightarrow$ the error could be large.

Intuition of Conditional number

$$A = U \Sigma V^T$$

rotation

rotation

Note that $A = U \Sigma V^T$

stretch: if conditional number is large $\Rightarrow$ stretch a lot along some direction
compress a lot in other directions

$$\Rightarrow A^T A = V \Sigma^T U^T U \Sigma V^T$$

$$= V \Sigma^T \Sigma V^T$$

$$\Rightarrow \sigma_{\max}(A^T A) = \sigma_{\max}^2(A)$$

$$\sigma_{\min}(A^T A) = \sigma_{\min}^2(A)$$

$$\Rightarrow \kappa(A^T A) = \kappa^2(A) \rightarrow A^T A \text{ Squaring the conditional number}$$

if A has big conditional number,

then working directly on $A^T A$ is not good.

Example: in linear regression, we work on

$$X^T X \beta = X^T y$$

if we do

$$\beta = (X^T X)^{-1} X^T y$$

We will lose a lot of precision if X has serious multicollinearity problem.

Since $X^T X$ squares the conditional number of X .

A computational safer approach for solving β : QR decomposition

$$X^T X \beta = X^T y$$

$$R^T Q^T Q R \beta = R^T Q^T y$$

$$R^T R \beta = R^T Q^T y$$

$$R \beta = Q^T y$$

Solve by backward substitution

QR decomposition and Gram-Schmidt Process

In order to understand QR decomposition, we first review the concept of projection

project a vector a into another vector b

the part projected
to b



$$(a - \lambda b)^T b = 0$$

$$\Rightarrow a^T b = \lambda b^T b$$

$$\Rightarrow \lambda = \frac{a^T b}{b^T b} = \frac{b^T a}{b^T b}$$

$$\Rightarrow \lambda b = \frac{b b^T}{b^T b} a$$

↓ projection matrix of
 b

project a vector a into another space B with basis $\{b_1, \dots, b_n\}$

$$(a - Bx)^T b_1 = 0$$

⋮

$$\Rightarrow B^T(a - Bx) = 0$$

$$(a - Bx)^T b_n = 0$$

$$\Rightarrow B^T a - B^T B x = 0$$

$$\Rightarrow x = (B^T B)^{-1} B^T a$$

$$\Rightarrow Bx = \underline{B(B^T B)^{-1} B^T} a$$

→ projection matrix into $C(B)$

Note that the projection matrix $P = A(A^T A)^{-1} A^T$ has the same column space as A .

$$\text{Since } P = A(A^T A)^{-1} A^T \Rightarrow C(P) \subseteq C(A)$$

$$PA = A \Rightarrow C(A) \subseteq C(P)$$

$$\text{So } C(A) = C(P)$$

Idea of Gram-Schmidt processes: Subtract from every new vector its projections in the direction already set.

Example: perform Gram-Schmidt on (a, b, c)

$$A = a$$

$$A' = \frac{A}{\|A\|}$$

$$B = b - \frac{AA^T}{A^T A} b = b - \frac{A^T b}{A^T A} A$$

$$B' = \frac{B}{\|B\|}$$

$$C = c - \frac{AA^T}{A^T A} c - \frac{BB^T}{B^T B} c = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B$$

$$C' = \frac{C}{\|C\|}$$

Other applications of QR decomposition